Toward Scalable Bayesian Uncertainty for Machine Learning with the Ray Tracing Sampler

Yasir Alsugair, **Joshua Speagle**

yasiralsugair@gmail.com, j.speagle@utoronto.ca

Abstract

This second report closes the convergence question that Report 1 opened and asks what a converged posterior is actually good for. Both architectures now formally pass our two-curve stationarity rule under the paper's noise-softened accept test: the MLP at 5.6 million minibatch steps and the CNN at 14.6 million, making them the study's only converged chains (the exact-test chains never passed). Convergence is not cosmetic: the CNN's test NLL moved from 0.0715 to 0.0854 during the final approach to its shell, so numbers read off an in-transit chain genuinely mislead. The converged CNN posterior reaches 98.92% accuracy with the best expected calibration error in our comparison (0.0033), essentially matching a 10-seed deep ensemble's accuracy (99.04%) from a single training run plus sampling. On uncertainty, at a matched review budget of 11% of the test set, the chain's per-sample disagreement flags 90% of its own errors (75x enrichment over unflagged images), between the point estimate (86%) and the ensemble (92%), and the three methods flag substantially different images (Jaccard overlap 0.32 to 0.52). On out-of-distribution probes (Gaussian noise, Fashion-MNIST, inverted MNIST) the chain's disagreement is the most consistent detector, flagging 88 to 98% of fakes even where its averaged confidence stays too high. Finally, testing our mentor's suggestion, we show that burn-in can be pre-paid: a prior-aware MAP optimizer started from a prior draw repairs the data fit far faster than it sheds weight norm, and chains launched from its transit states accept 92 to 96% at the full tuned step size, where raw bad-fit starts had needed 12x to 350x smaller steps. The practical recipe: stop the optimizer when the weight norm first nears the chain's settle level, then sample.

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Code: https://github.com/YasirAlsugair/ray-tracing-sampler-experiments

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1 Introduction

Report 1 built the testbed: small MNIST classifiers (an MLP with 50,890 parameters, a CNN with 12,810) whose loss-to-posterior relationship is derived exactly, so the sampled distribution is the Bayes posterior at $T = 1$ by construction and every approximate chain can be scored against an exact Metropolis chain. Its headline findings were that exact chains hold 97 to 99% acceptance while remaining far from stationary (the posterior lives at large weight norms and the chain must walk there), that the accept test cannot run on minibatch estimates, and that the paper's noise-softened test (Eq. 33 of [1]) had carried the MLP to the study's only formal convergence.

This week asked three questions. Does the CNN converge too, and does convergence change its answers? Is the converged posterior's uncertainty actually useful next to deep ensembles [2] and MC dropout [3], measured on equal terms? And, following a suggestion from our mentor, can the long burn-in be pre-paid by an optimizer that knows about the prior?

2 Methods update

Everything from Report 1 stands; four things are new. First, the CNN convergence program: the recipe-chosen Eq. 33 setting ($dt = 10^{-4}$, refresh 5) was extended in 250k-step legs, each leg checkpointed, until the two-curve drift rule (data misfit and weight norm must both level) rendered a verdict. Second, a matched-budget protocol for comparing uncertainty: each method's flagging threshold is set so it marks the same 11% of the real test set, and we then measure what fraction of the method's own test errors fall inside its flagged set. Third, an out-of-distribution probe: Gaussian noise, Fashion-MNIST [4], and inverted MNIST images are scored at that same shared threshold, asking what fraction of fakes each method flags as uncertain. Fourth, the MAP-transit protocol: we run a MAP optimizer on the full log posterior (likelihood plus prior) starting *from a prior draw*, snapshot its trajectory at several epochs, and launch exact Metropolis chains (2,000 trajectories, the full tuned step $dt = 3.5 \times 10^{-4}$, $L = 30$) from each snapshot; these runs used a cloud GPU, with chains of 2,000 trajectories taking about three minutes each.

3 Results

3.1 The CNN converges, and convergence matters

The CNN passed the two-curve rule at 14.6 million minibatch steps (Figure 1). Its weight norm settles at a final-quarter mean of 12,211, about 95% of its shell at $D = 12{,}810$, where the MLP had settled at 97% of its own shell: how far below D a posterior settles is architecture dependent, consistent with the likelihood constraining a different fraction of directions in a convolutional network. The training misfit's running median settles near 350 nats. Two observations sharpen earlier conclusions (Figure 2). The gate's rejection rate trended in opposite directions on the two architectures as the chains settled (MLP 51% down to 27%, CNN 37% up to 55%), so the Eq. 33 noise floor is state and architecture dependent, not a constant of the method. And the CNN's test NLL moved from 0.0715 to 0.0854 during the final norm creep alone: an in-transit chain does not just add noise to the answer, it reports a systematically different answer.

The converged CNN predictive reaches accuracy 0.9892 with NLL 0.0854 and ECE 0.0033 (Figure 3). Against the baselines (point estimate 0.9847 / 0.0474 / 0.0020; 10-seed ensemble 0.9904 / 0.0310 / 0.0071) the converged posterior essentially matches ensemble accuracy from one training run plus sampling, posts the best calibration of any sampled method, and still pays an NLL premium, the same cold-posterior signature the MLP showed.

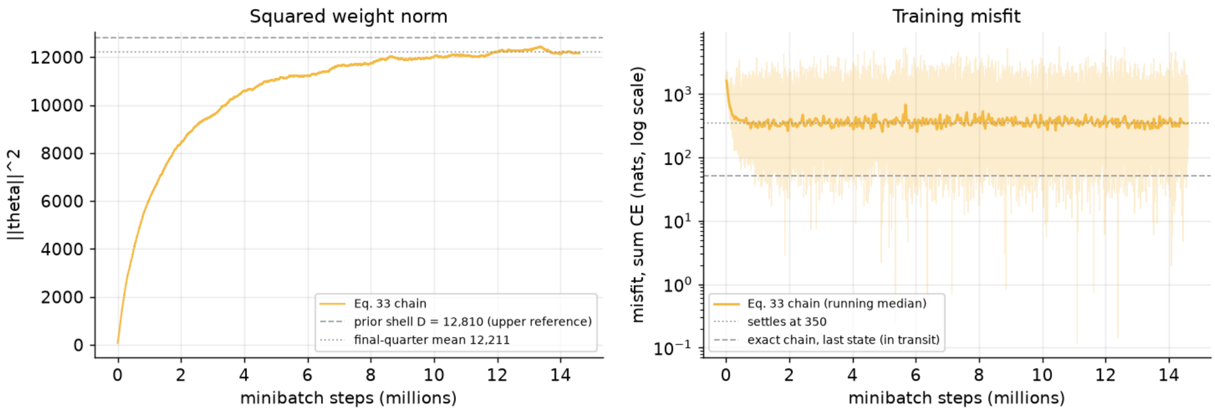

Figure 1 The CNN Eq. 33 chain to convergence. Left: squared weight norm against the prior shell (D = 12,810, dashed) and the final-quarter mean (12,211, dotted). Right: training misfit (running median), settling near 350 nats; the exact chain's last in-transit state is shown for reference.

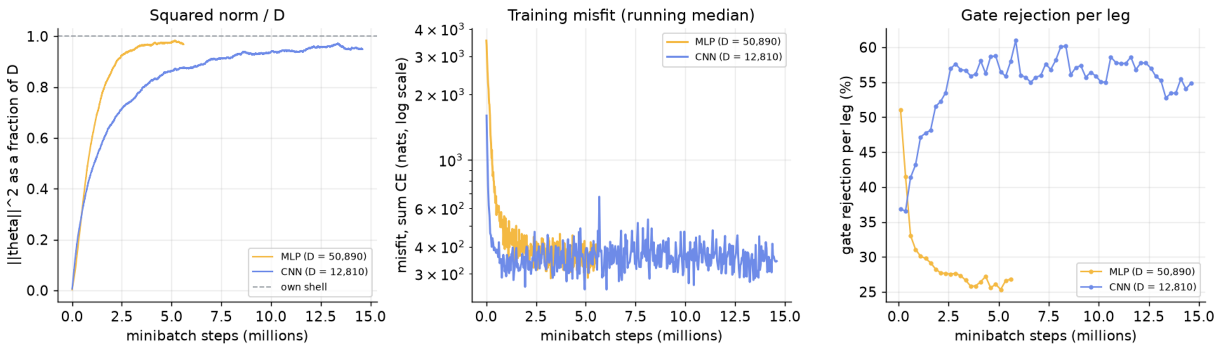

Figure 2 Both architectures side by side. Left: squared norm as a fraction of each network's own shell (MLP settles at 97%, CNN at 95%). Middle: training misfit. Right: gate rejection per leg, trending down for the MLP and up for the CNN.

3.2 What the posterior's uncertainty buys

At the matched 11% review budget, the converged chain's per-sample disagreement catches 90% of its own test errors at 75x enrichment over the unflagged images; the point estimate's confidence catches 86% at 52x, and the ensemble 92% at 93x. The more interesting number is the overlap: the flagged sets agree only partially (Jaccard 0.32 to 0.52), so the methods disagree about which images are dangerous and are not interchangeable instruments. On the out-of-distribution probes (Figure 4) the chain is the most consistent detector, flagging 98% of Gaussian noise, 88% of Fashion-MNIST, and 93% of inverted MNIST, even in regimes where its averaged confidence remains too high: confidence and disagreement decouple, and the disagreement is the more reliable signal. A member-scaling check adds one more contrast: two ensemble members already deliver most of the ensemble's benefit, while the chain's predictive is still improving at 50 samples.

3.3 Pre-paying burn-in: MAP-transit starts

Report 1 showed that burn-in has two unequal jobs (fit the data, walk to the shell) and that bad-fit starts force tiny step sizes. Our mentor suggested the constructive version: run a prior-aware optimizer and start the chain from its *transit*. A MAP optimizer on the full log posterior, started from a prior draw, repairs the fit hundreds of times faster than it sheds norm (Table 1), so its early epochs pass through exactly the states a sampler wants to start from: good fit, shell-level norm.

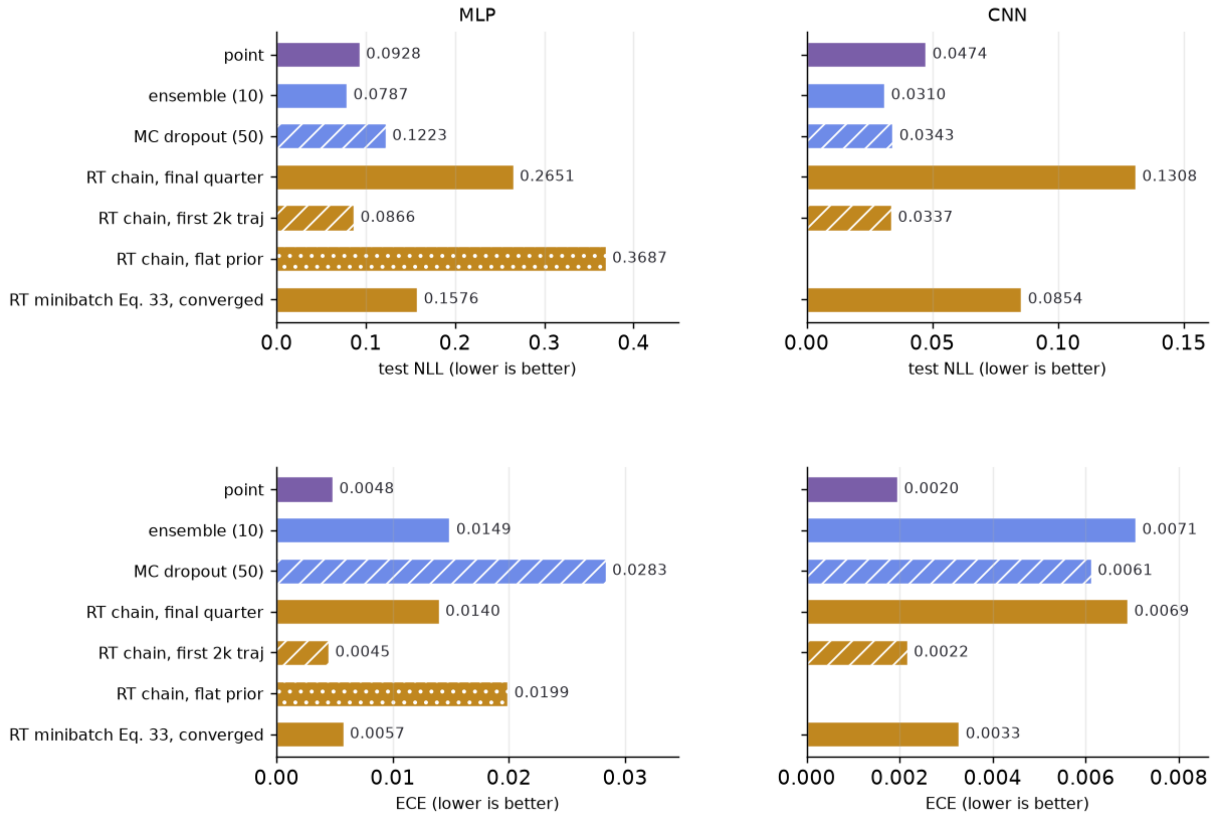

Figure 3 Test NLL (top) and expected calibration error (bottom) for every method on both architectures. The converged Eq. 33 rows are the only entries from formally stationary chains.

Chains from epochs 10 to 20 accept 92 to 96% at the full tuned step and start at the shell, staying there while dropping tens of thousands of nats of misfit within 2,000 trajectories; the Adam-start chain of Report 1 needed 20,000 trajectories for its walk. By epoch 50 the optimizer has slid off the shell (norm 21,295) and the walk is owed again. The recipe is therefore to stop early: halt the optimizer when the weight norm first nears the chain's settle level, then sample. Burn-in becomes something an optimizer pre-pays in minutes.

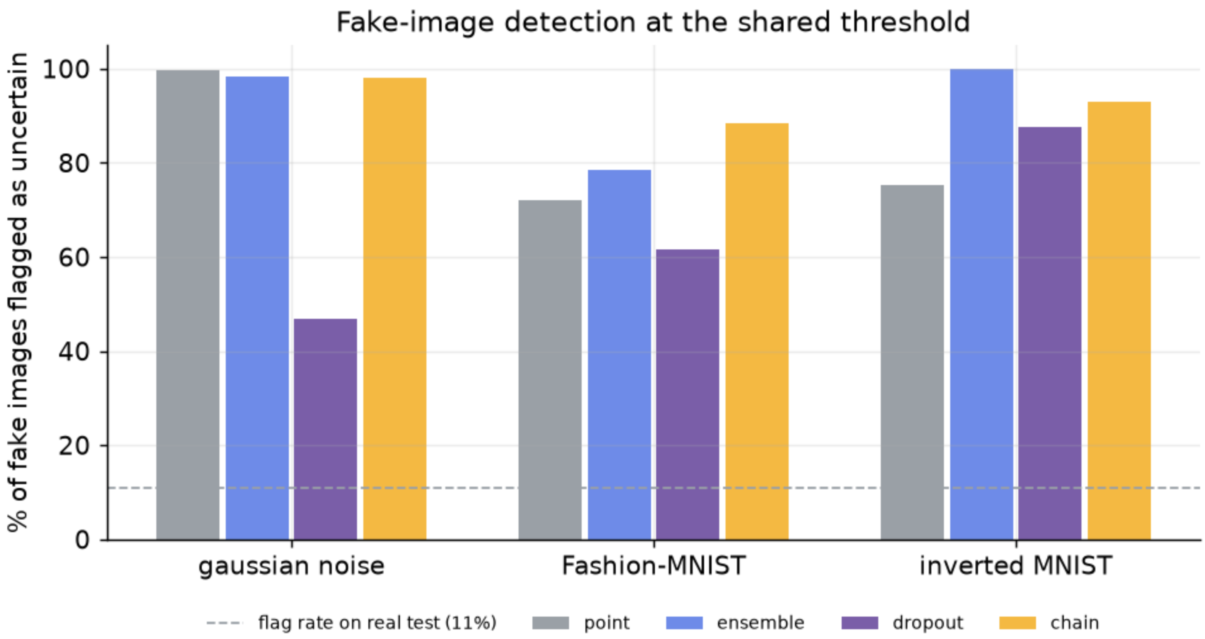

Figure 4 Fraction of fake images flagged as uncertain at the shared threshold (11% flag rate on real test images) for three out-of-distribution sets. The chain's disagreement is the most consistent detector across all three.

Table 1 Chains launched from MAP transit states (2,000 trajectories each at the full tuned step). Raw bad-fit starts in Report 1 needed 12x to 350x smaller steps; every transit start accepts at full step.

start	misfit_0	$\|\theta\|_0^2$	acceptance	misfit_{2k}	$\|\theta\|_{2k}^2$
MAP epoch 10	58,740	49,401	92%	7,537	49,203
MAP epoch 15	27,668	48,027	96%	7,363	48,407
MAP epoch 20	17,771	45,453	93%	7,269	46,264
MAP epoch 50	4,985	21,295	97%	4,118	26,858

4 Discussion

Three lessons carry the week. First, the noise-softened test's time-tax-not-bias story now holds on both architectures, and it delivered the only stationary chains in the study; the CNN's NLL drift during its final approach shows what the stationarity pedantry is protecting against. Second, confidence and disagreement are different instruments: the averaged predictive can stay overconfident on fakes that the sample-to-sample disagreement flags immediately, and different uncertainty methods flag substantially different images, which argues for reporting what a method flags, not just how much. Third, the two-jobs picture of burn-in is now actionable: since fit difficulty sets the step size, pre-paying the fit with a prior-aware optimizer, stopped in transit, removes the expensive job entirely.

5 Limitations and next steps

The scope limits from Report 1 stand: MNIST scale, single chains exploring single basins, and effective-sample-size estimates that remain upper bounds. The MAP-transit result is one architecture and one prior so far, and the stop-early rule references the chain's settle level, which is known here from ground truth and must be estimated in the wild. Next steps stay on the announced track: use the misfit and norm curves as auto-tuning signals, study the temperature and prior-scale dials on the converged setup (implemented so the accept test stays consistent with the tempered target), extend the MAP-transit recipe to the CNN, and bring additional samplers onto the same harness for like-for-like comparison.

6 Conclusion

Both posteriors in this study are now formally converged, something the exact test never achieved here, and the converged CNN posterior turns out to be genuinely competitive: ensemble-level accuracy, the best calibration among sampled methods, and the most consistent out-of-distribution flag in our comparison, from one training run plus sampling. Convergence changed the answers, uncertainty proved to be the posterior's real product, and burn-in, the cost that dominated Report 1, now looks like something an optimizer can pre-pay. The repository with every run, figure, and the executed notebook remains public.

References

[1] Peter Behroozi. "The Ray Tracing Sampler". In: *arXiv preprint arXiv:2510.25824* (2025).

[2] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. "Simple and scalable predictive uncertainty estimation using deep ensembles". In: *Advances in Neural Information Processing Systems*. 2017.

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[4] Han Xiao, Kashif Rasul, and Roland Vollgraf. "Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms". In: *arXiv preprint arXiv:1708.07747* (2017).